Pythagoras' Theorem

Name _____ Date _____ Calculator allowed Version 1

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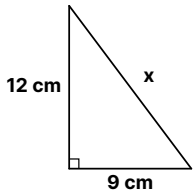
KEY WORDS right-angled triangle *(tapatoru hāngai)* · hypotenuse *(tapa roa)* · square root *(pūtakerua)*

A Finding the hypotenuse ●○○

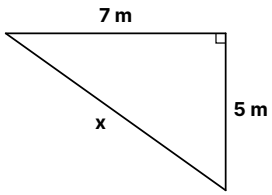
Remember: $a^2 + b^2 = c^2$, where c is the hypotenuse, the longest side. Round to 1 decimal place where needed.

1. Find the hypotenuse, x. Diagrams are not to scale.

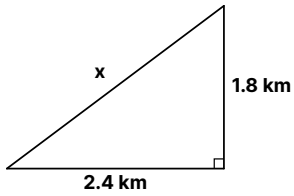
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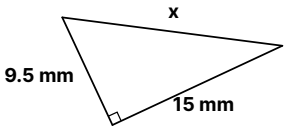
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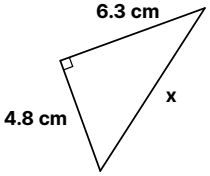
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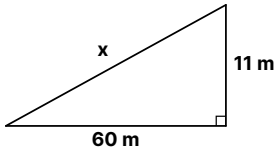
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f)

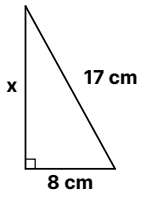


B Finding a shorter side ●●○

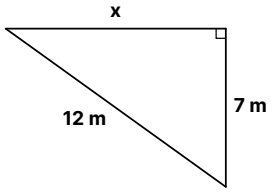
Watch out: When the hypotenuse is given, subtract: $a^2 = c^2 - b^2$.

2. Find the length of the side marked x.

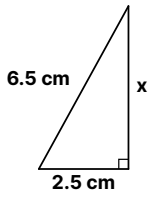
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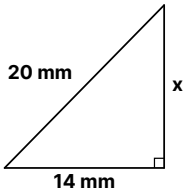
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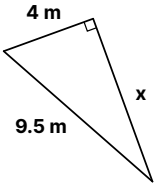
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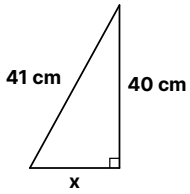
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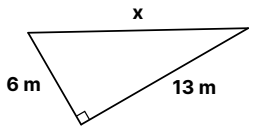


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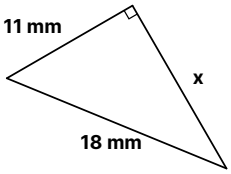
C Mixed practice ●●○

3. Find x. First decide whether x is the hypotenuse or a shorter side.

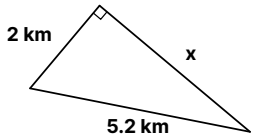
a)



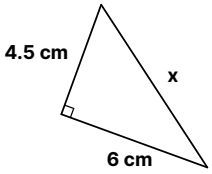
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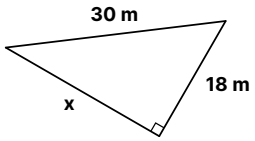
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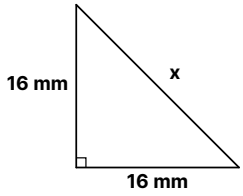
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e)

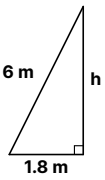


f)



D Problems in context ●●●

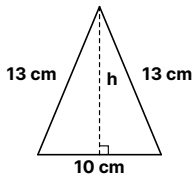
4. A 6 m ladder leans against a wall. Its foot is 1.8 m from the wall. How high up the wall does the ladder reach?



5. A television screen is 88 cm wide and 50 cm high. Find the length of its diagonal, to the nearest centimetre.

6. A boat sails 12 km north and then 9 km east. How far is it from where it started?

7. An isosceles triangle has two sides of 13 cm and a base of 10 cm. Find its height.



8. A wheelchair ramp covers 4 m of ground and rises 0.6 m. Find the length of the sloping surface of the ramp, to 2 decimal places.

E Distance between two points ●●●

> **Remember:** Think of a right-angled triangle. Its shorter sides are the difference in the x values and the difference in the y values.

9. Find the distance between each pair of points. Round to 1 decimal place where needed.

a) $(1,\ 2)$ and $(4,\ 6)$

b) $(0,\ 0)$ and $(5,\ 12)$

c) $(-2,\ 1)$ and $(4,\ 9)$

d) $(2,\ 3)$ and $(5,\ 8)$

e) $(-3,\ -1)$ and $(2,\ 4)$

f) $(-4,\ 5)$ and $(3,\ -2)$

★ Extension

10. A box is 12 cm long, 4 cm wide and 3 cm high. Find the length of the longest straight rod that fits inside it, from one corner to the opposite corner.

11. The diagonal of a square is 10 cm long. Find the length of its sides, to 1 decimal place.

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Pythagoras' Theorem

Name _____ Date _____ Calculator allowed Version 2

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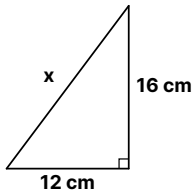
KEY WORDS right-angled triangle *(tapatoru hāngai)* · hypotenuse *(tapa roa)* · square root *(pūtakerua)*

A Finding the hypotenuse ●○○

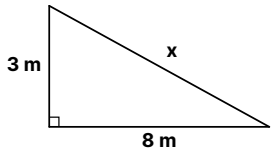
Remember: $a^2 + b^2 = c^2$, where c is the hypotenuse, the longest side. Round to 1 decimal place where needed.

1. Find the hypotenuse, x. Diagrams are not to scale.

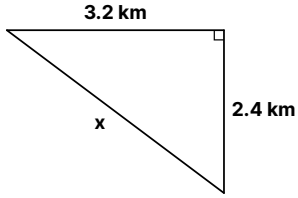
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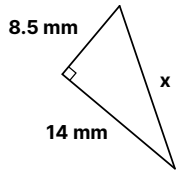
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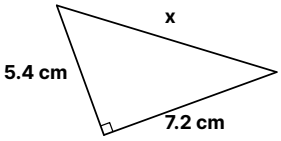
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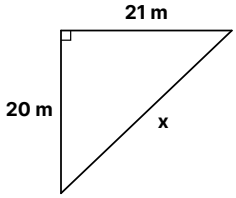
d)



e)



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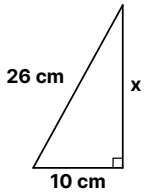


B Finding a shorter side ●●○

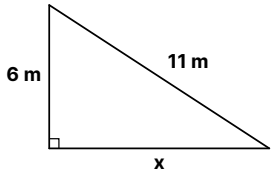
Watch out: When the hypotenuse is given, subtract: $a^2 = c^2 - b^2$.

2. Find the length of the side marked x.

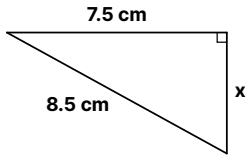
a)



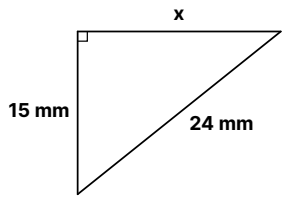
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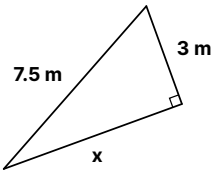
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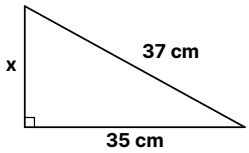
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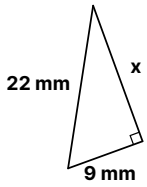


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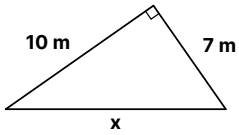
C Mixed practice ●●○

3. Find x. First decide whether x is the hypotenuse or a shorter side.

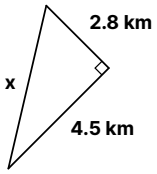
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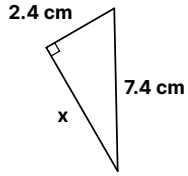
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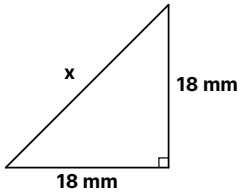
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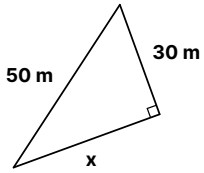
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e)

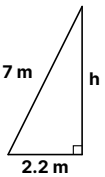


f)



D Problems in context ●●●

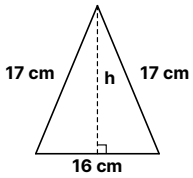
4. A 7 m ladder leans against a wall. Its foot is 2.2 m from the wall. How high up the wall does the ladder reach?



5. A television screen is 120 cm wide and 68 cm high. Find the length of its diagonal, to the nearest centimetre.

6. A boat sails 15 km north and then 8 km east. How far is it from where it started?

7. An isosceles triangle has two sides of 17 cm and a base of 16 cm. Find its height.



8. A wheelchair ramp covers 5 m of ground and rises 0.75 m. Find the length of the sloping surface of the ramp, to 2 decimal places.

E Distance between two points ●●●

> **Remember:** Think of a right-angled triangle. Its shorter sides are the difference in the x values and the difference in the y values.

9. Find the distance between each pair of points. Round to 1 decimal place where needed.

a) $(2,\ 1)$ and $(5,\ 5)$ **b)** $(0,\ 0)$ and $(8,\ 15)$ **c)** $(-1,\ 2)$ and $(5,\ 10)$

d) $(1,\ 4)$ and $(6,\ 7)$ **e)** $(-2,\ -3)$ and $(4,\ 3)$ **f)** $(-5,\ 4)$ and $(2,\ -1)$

★ Extension

10. A box is 8 cm long, 4 cm wide and 1 cm high. Find the length of the longest straight rod that fits inside it, from one corner to the opposite corner.

11. The diagonal of a square is 14 cm long. Find the length of its sides, to 1 decimal place.

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